SHORT ANSWER TEST — SET - 1

50/50

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1) Simplex channel = 25 kHz
Total spectrum = 45 MHz
Full duplex needs 2 simplex channels

=> 45 MHz × 1000 = 45000 kHz

Simplex channels are = $\frac{45000}{25}$ = 1800

2 Full duplex channels = $\frac{1800}{2}$ = 900 ~ 2

∴ 900 full duplex voice channels are there.

3) Carrier spacing set at = 200 kHz
Carriers = 100
Time slots for carriers = 8 2

=) 100 × 8 = 800

∴ 800 simultaneous voice channels

5) Carrier = -60 dBm

Co-channel interference = -92 dBm

Carrier to Interference Ratio (C/I),

$$\frac{C}{I} = -60 - (-92)$$

$$= -60 + 92$$

$$= \underline{32 \text{ dB}} \quad 2$$

7) Path loss exponent = n = 4

Power = $P \propto \frac{1}{d^4}$

$$\Rightarrow P_{new} = 64 / 2^4$$

$$= \frac{64}{16} = \underline{4W} \quad 2$$

∴ The power recieved is 4W.

9) Cluster size = N = 4

No. of voice channels = 15

Total channel count = cluster size x no. of voice channels

$$= 4 \times 15$$

$$= \underline{60} \quad 2$$

The total no. of channels are 60.

TDMA frame time = 5 ms

No. of slots = 8

Duration of single slot = $\frac{5}{8}$

= 0.625 ms

∴ Duration of a single slot is 0.625 ms. 2

13) Power transmitted = 8 W

Path loss = 33 dB

P(dBm) = $10 \log_{10}(8000)$ (W to dBm)

= 39.03 dBm

Path loss = 33 dB

P = 39.03 − 33

= 6.03 dBm 2

15) No. of cells = 6

No. of simultaneous calls = 250

Call handling capacity = 6 × 250

= 1500

∴ 1500 simultaneous calls. 2

17) No. of channels = 150

For each channel : 25 kHz

Bandwidth = 150 × 25

= 3750 kHz ✓

= 3.750 MHz

19) Vehicle speed = 90 km/h

Cell radius = 1.5 km

Distance traveled in 1hr = 90 km

Diameter = 2R = 2 × 1.5 = 3 km

⇒ Handoffs = $\frac{90}{3}$ = 30 ✓

∴ 30 handoffs per hour.

21) Area = 400 sq km

Radius = 400 m = $\frac{4}{1000}$ = 0.4 km

Approx. cell area = R^2

⇒ $R^2 = 0.4^2 = 0.16$ km^2 ✓

No. of cells = $\frac{400}{0.16}$ = 2500 cells

∴ Approximately 2500 cells.

No. of subscribers = 1200

No. of calls per hour = 4

Time per call = 2.5 minutes = $\frac{2.5}{60}$ hr

$$\Rightarrow 1200 \times 4 \times \frac{2.5}{60}$$

$$= 1200 \times 4 \times 0.04166$$

$$= \underline{200 \text{ Erlangs}} \quad 2$$

25) No. of simultaneous users = 600

No. of stations = 1200

$$\Rightarrow 600 \times 1200 = \underline{720000 \text{ users}} \quad 2$$

27) Call attempts = 40000/min

Time = 3min

Traffic = $40000 \times 3 = 120000$ Erlangs

Blocked = $120000 - 100000 = \underline{20000}$

Fraction = $\frac{20000}{120000} \times 100 = \underline{16.67\%}$ 2

29) Current load = 9000 calls/sec

$$\Rightarrow \frac{9000}{12000} = \underline{0.75} \times 100 = \underline{75\%} \quad 2$$

31) 30 MHz = 30000 kHz

⇒ $\frac{30000}{200}$ = 150 channels 2

33) Power = 12w

⇒ $10 \log_{10}(12000)$ = 40.79 dBm

Path loss = 45 dB

40.75 − 45 = −4.21 dBm 2

35) Frame = 4.8 ms

Slots = 8 2

⇒ $\frac{4.8}{8}$ = 0.6 ms

37) Call rate = 70000 /min

⇒ 70000 × 4 = 280000

Capacity = 250000

⇒ 280000 − 250000 = 30000

⇒ $\frac{30000}{280000} \times 100$ = 10.71 % 2

39) Total calls = 60000, DP = 0.4 %

⇒ 60000 × 0.004 % = 240 calls / sec

2

Area = 2.5 km^2

No. of stations = 150

⇒ 2.5 × 150 = 375 km^2 2

43) No. of users = 6000

Total subscribers = 150,000

⇒ $\frac{150000}{6000}$ = 25 paging cycles 2

45) Signal = −85 dBm

Noise = −115 dBm

SNR = −85−(−115) 2

= 30 dB

47) Subscribers = 1500

Calls = 4

Duration = 2.5 min

⇒ $1500 \times 4 \times \frac{2.5}{60}$

= 250 Erlangs 2

49) Connections : 1500

Rate / connection : 600 kbps

=) $1500 \times 600 = 900\,000$ kbps

$= \underline{900}$ Mbps

∴ The throughput is $\underline{900}$ Mbps. 2